[W9] Midterm Review and Board Notes

Advanced Machine Learning for Artificial Intelligence · October 27, 2026

Use the writing areas to record the board work and your own calculations. The original exam contains five questions worth 20 points each, with three parts per question (100 points in total).

Show working. Under the original marking policy, an entire question with no working earns no credit. With partial working, marks are awarded for the steps actually demonstrated. Consistent follow-through of an earlier arithmetic error is distinguished from a new conceptual error. Equivalent expressions and the original rounding allowances remain valid.

Marking is planned to finish this week. Results will be announced after marking is complete.

Q1(a) · Exact Inference on a Small Chain · 8 points

Let $A \to B \to C$ be a binary chain, with states $0, 1$ and $p(A) = (0.5, 0.5)$. The transition tables are

$$T_{AB} = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}, \qquad T_{BC} = \begin{pmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{pmatrix}.$$

Rows index the source state; columns index the destination state. A fixed observation e depends only on C, with likelihoods $(p(e \mid C = 0), p(e \mid C = 1)) = (0.2, 1)$. Keep messages unnormalized.

Compute the left message $m_L(b) = p(B = b)$ and the right message $m_R(b) = p(e \mid B = b)$ for $b = 0, 1$. Show the state-indexed sums.

Original marking criteria

Assessment item	Credit conditions	Points
Left-message sums	Correct state-indexed setup; partial credit for one correctly set-up state.	2
Left-message values	One point per state; arithmetic and consistent follow-through assessed separately.	2
Right-message sums	Correct transition row and evidence likelihood for each B state.	2
Right-message values	One point per original unnormalized value; original values must remain identifiable.	2

Q1(a) · Board notes and calculations

Q1(b) · Exact Inference on a Small Chain · 6 points

Use the same setting as Q1(a).

Use the messages to compute the weights $w_b = p(B = b, e)$ and the evidence $p(e)$.

Original marking criteria

Assessment item	Credit conditions	Points
Message combination	Correct same-state combination for both states; partial credit for one state.	2
Joint weights	One point per state; consistent arithmetic follow-through is recognized.	2
Evidence	Correct aggregation setup and calculation, one point each.	2

Q1(b) · Board notes and calculations

Q1(c) · Exact Inference on a Small Chain · 6 points

Use the same setting as Q1(a).

Compute $p(B = 0 \mid e)$ and $p(B = 1 \mid e)$, then verify that they sum to one.

Original marking criteria

Assessment item	Credit conditions	Points
Posterior for state 0	Common normalization and resulting value, one point each.	2
Posterior for state 1	Common normalization and resulting value, one point each; equivalent complement calculation accepted.	2
Normalization check	Explicit addition and valid verification, one point each; stating the property alone earns at most one point.	2

Q1(c) · Board notes and calculations

Q2(a) · Mean-Field Variational Inference · 8 points

The exact posterior is $p(z \mid x) = \mathcal{N}(\mu, P^{-1})$. Approximate it by $q(z) = \mathcal{N}(m, \mathrm{diag}(v_1, v_2))$, where

$$P = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \mu = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad m^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Use the Gaussian CAVI updates

$$v_j = \frac{1}{P_{jj}}, \qquad m_j \leftarrow \mu_j - \frac{P_{jk}}{P_{jj}}(m_k - \mu_k), \quad k \neq j.$$

All logarithms are natural. Show the substitutions and intermediate calculations.

Compute $C = P^{-1}$, the two exact marginal variances, the optimal diagonal variances v_1^*, v_2^*, and each shortfall $C_{jj} - v_j^*$.

Original marking criteria

Assessment item	Credit conditions	Points
Determinant and inverse setup	Correct determinant or equivalent elimination; method credit retained for a single arithmetic slip.	2
Covariance and marginal variances	Correct matrix structure and identification of its diagonal; consistent arithmetic follow-through accepted.	2
Optimal diagonal variances	One point per coordinate: justified substitution and calculation.	2
Variance shortfalls	One point per coordinate: requested subtraction direction and calculation.	2

Q2(a) · Board notes and calculations

Q2(b) · Mean-Field Variational Inference · 8 points

Use the same setting as Q2(a).

Starting from $m^{(0)}$, perform one sequential sweep: update coordinate 1, then coordinate 2 using the new coordinate 1. Report both updated means.

Original marking criteria

Assessment item	Credit conditions	Points
First-coordinate substitution	Use the supplied update with the initial second coordinate; partial method credit where applicable.	2
First-coordinate calculation	Show the calculation; distinguish a new arithmetic slip from an already assessed error.	2
Sequential input to coordinate 2	Use the newly updated first coordinate; consistent follow-through accepted.	2
Second-coordinate calculation	Assess subsequent arithmetic separately from the choice of sequential input.	2

Q2(b) · Board notes and calculations

Q2(c) · Mean-Field Variational Inference · 4 points

Use the same setting as Q2(a).

For this model, $\log p(x) = -3.0000$, the exact diagonal-family optimum has $\mathcal{L}^* = -3.1438$, and after (b), with the variances in (a), $\mathcal{L}_{\text{current}} = -3.3313$ (rounded to four decimal places). Compute the approximation gap and the remaining optimization gap, in nats.

Original marking criteria

Assessment item	Credit conditions	Points
Approximation gap	Select the correct pair of levels and calculate their difference, one point each.	2
Optimization gap	Select the correct pair of levels and calculate their difference, one point each.	2

Q2(c) · Board notes and calculations

Q3(a) · Importance Sampling and Effective Sample Size · 8 points

For $f(z) = z$, the normalized target distribution p and proposal distribution q are:

z	0	1	2
$p(z)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
$q(z)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

Four independent draws from q are $(0, 1, 1, 2)$; count repeated values separately. Use

$$w_i = \frac{p(z^{(i)})}{q(z^{(i)})}, \quad W_i = \frac{w_i}{\sum_{j=1}^{4} w_j}, \quad N_{\text{eff}}^{\text{weight}} = \frac{1}{\sum_{i=1}^{4} W_i^2}.$$

Compute the raw weights w_i, their sum, and normalized weights W_i, in draw order.

Original marking criteria

Assessment item	Credit conditions	Points
Four raw weights	One point per draw: half for the ratio substitution and half for arithmetic; repeated draws retain separate slots.	4
Weight sum	Add all four contributions; method credit retained for a new arithmetic slip.	1
Normalization denominator	Use the sum of all raw draw weights.	1
Four normalized weights	Half a point per draw in the original order; consistent follow-through accepted.	2

Q3(a) · Board notes and calculations

Q3(b) · Importance Sampling and Effective Sample Size · 8 points

Use the same setting as Q3(a).

Compute the exact target expectation $\mathbb{E}_p[z]$, the direct IS estimate, and the self-normalized IS (SNIS) estimate from these four draws. Show the denominator of each estimate.

Original marking criteria

Assessment item	Credit conditions	Points
Exact target expectation	State-space setup and result, one point each.	2
Direct IS	Weighted numerator, requested denominator, and arithmetic, one point each.	3
SNIS	Weighted numerator, correct normalization, and arithmetic, one point each; equivalent normalized-weight form accepted.	3

Q3(b) · Board notes and calculations

Q3(c) · Importance Sampling and Effective Sample Size · 4 points

Use the same setting as Q3(a).

Compute the weight ESS using the normalized weights from (a). Give both an exact fraction and a decimal value.

Original marking criteria

Assessment item	Credit conditions	Points
Squared-weight sum	Correct four-draw setup and calculation, one point each; equivalent raw-weight ESS formula accepted.	2
ESS	Reciprocal operation and result, one point each; the result mark requires both fraction and decimal, with half for one correct form.	2

Q3(c) · Board notes and calculations

Q4(a) · Predictive Loss and Paired Comparisons · 6 points

Three responses are conditionally independent, with $y = (0, 2, 4)$, predicted means $\mu_A = (1, 1, 3)$ and $\mu_B = (0, 1, 4)$. Both models use the same known variance $\sigma^2 = 2$:

$$p(y_i \mid \mu_i, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \mu_i)^2}{2\sigma^2}\right].$$

For (c), use these separate experiment records. Match conditions by run ID, not row position. Lower error is better.

A: run ID	A: error	B: run ID	B: error
1	0.4	3	0.2
2	0.3	1	0.3
3	0.5	2	0.4

Derive the summed Gaussian NLL in terms of $\mathrm{SSE} = \sum_{i=1}^{3} (y_i - \mu_i)^2$, retaining the normalizing constant. Use natural logarithms.

Original marking criteria

Assessment item	Credit conditions	Points
Independent joint density	Show the product over all observations; partial credit for incomplete product or independence stated alone.	2
Negative log and quadratic term	Credit the log transformation and the supplied-variance coefficient separately.	2
Normalizing constant	Show its general structure and substitute the supplied sample size and variance.	2

Q4(a) · Board notes and calculations

Q4(b) · Predictive Loss and Paired Comparisons · 8 points

Use the same setting as Q4(a).

Calculate SSE and MSE for each model, and $\mathbf{NLL}_{A} - \mathbf{NLL}_{B}$. Show residual calculations.

Original marking criteria

Assessment item	Credit conditions	Points
Model A residuals and SSE	Credit the residual/squared-residual calculation and the sum separately; consistent opposite residual sign accepted.	2
Model A MSE	Divide the SSE by the number of observations; consistent follow-through accepted.	1
Model B residuals and SSE	Credit the residual/squared-residual calculation and the sum separately; consistent opposite residual sign accepted.	2
Model B MSE	Divide the SSE by the number of observations; consistent follow-through accepted.	1
Signed NLL difference	Correct common-term cancellation and calculation in the requested A-minus-B direction.	2

Q4(b) · Board notes and calculations

Q4(c) · Predictive Loss and Paired Comparisons · 6 points

Use the same setting as Q4(a).

Match by run ID. Calculate $d_i = e_{A,i} - e_{B,i}$ for runs $1, 2, 3$, their mean, and the number of runs B wins. A tie is not a win.

Original marking criteria

Assessment item	Credit conditions	Points
Matched difference, run 1	Pair the records with the same run ID and retain the specified subtraction direction.	1
Matched difference, run 2	Pair the records with the same run ID and retain the specified subtraction direction.	1
Matched difference, run 3	Pair the records with the same run ID and retain the specified subtraction direction.	1
Paired mean	Correct aggregation and calculation, one point each; an equivalent difference of means is accepted.	2
B win count	Count differences with the correct sign; ties are not wins.	1

Q4(c) · Board notes and calculations

Q5(a) · Bayesian Prediction in Weight and Function Space · 8 points

Let $f(x) = xw$, with $w \sim \mathcal{N}(0, \frac{1}{2})$ and $y_i = x_i w + \epsilon_i$, where the training noises $\epsilon_i \sim \mathcal{N}(0, 1)$ are independent of each other and of w. The second argument of $\mathcal{N}$ is the **variance**. Observe $D = \{(1, 1), (2, 3)\}$.

Find the posterior mean and variance of $w \mid D$. Show the prior–likelihood calculation, using either square completion or a precision update.

Original marking criteria

Assessment item	Credit conditions	Points
Prior and both likelihoods	Credit the prior-variance contribution and both data contributions separately.	2
Quadratic or precision calculation	Correct quadratic and linear coefficients, or equivalent precision and information contributions.	2
Posterior mean	Identify the center or equivalent precision relation and show the resulting calculation.	2
Posterior variance	Use the precision-to-variance relation and calculate the variance; distinguish standard deviation.	2

Q5(a) · Board notes and calculations

Q5(b) · Bayesian Prediction in Weight and Function Space · 8 points

Use the same setting as Q5(a).

For a **new response** at $x_* = 2$, let $f_* = 2w$ and $y_* = f_* + \epsilon_*$, where $\epsilon_* \sim \mathcal{N}(0, 1)$ is independent of w and the training noises. Calculate $E[f_* \mid D]$, $\mathbf{Var}(f_* \mid D)$ and $\mathbf{Var}(y_* \mid D)$. Show the variance calculations.

Original marking criteria

Assessment item	Credit conditions	Points
Latent-function mean	Correct expectation transformation and calculation, one point each.	2
Latent-function variance	Credit squared scaling, use of the posterior variance, and calculation separately.	3
New-response variance	Credit independent-noise addition, correct noise variance, and calculation separately.	3

Q5(b) · Board notes and calculations

Q5(c) · Bayesian Prediction in Weight and Function Space · 4 points

Use the same setting as Q5(a).

Before observing D, derive the prior kernel $k(x, x') = \mathrm{Cov}(f(x), f(x'))$ and calculate $k(-1, 2)$. Show how the weight prior determines this covariance.

Original marking criteria

Assessment item	Credit conditions	Points
Prior kernel derivation	Use the shared-weight covariance and its prior variance.	2
Prior covariance for the input pair	Show substitution of the requested input pair and the resulting calculation.	2

Q5(c) · Board notes and calculations

After-class review notes

Which step will you revisit? Record the relevant question label and your own explanation. This sheet does not introduce a new submission task.

References

Original instructor-provided midterm, October 20, 2026. Previous class materials: Course home.